Warcraft Card Duels – Individual Project

ASE 420

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Project Description

- Problem: Players want an engaging turn-based RPG experience with varied heroes, abilities, enemies, and challenging boss fights.
- Goal: Create a turn-based RPG game with a variety of heroes with unique abilities and enemies to face.
- Proposed Solution: Build a Python game where players create a team of 3 heroes and fight randomly generated groups of enemies before facing powerful bosses.

Core Features Overview

- Hero & Team System
- Combat System
- Encounters and Progression

Feature 1: Heros & Team System

Description

Players will be able to select from a variety of heros to create a team of 3

Requirements

- Maintain a substancial roster of unique heros
- Each hero has a unique class, specialization, and stats
- Each hero has 3-5 unique abilities
- Players can create a team of 3 heros to traverse dungeons and other encounters

Feature 2: Combat System

Description

Turn-based combat between a player's team and enemies

Requirements

- Determine turn order for each hero or enemy
- Allow each hero and enemy to use abilities
- Abilities have unique effects or damage
- Track health and other combat stats
- Detect victory or defeat conditions

Feature 3: Encounters and Progression

Description

Fight waves of enemies before a boss encounter

Requirements

- Each wave of enemies has 1-3 random mobs
- Enemies have different types and behaviors
- Scaling difficulty as game progresses
- Provide clear victory conditions

Total Features and Requirements

Features: 3

Requirements: 13

Architecture

- Modular, object-oriented architecture
- Separate systems for heroes, combat, encounters, and UI
- Designed to make adding new heroes, abilities, and enemies easier

Design

- Encapsulation — Classes manage their own data and behavior.
- Separation of Concerns — Each system has a specific responsibility.
- Inheritance — Shared character functionality can be reused.
- Composition — Heroes can contain their abilities.
- State Pattern — Used to manage different game states.
- Strategy Pattern — Can be used for different enemy behaviors.

Schedule & Milestones

Sprint 1

- Week 4: Setup
- Week 5: Game States
- Week 6: Characters
- Week 7: Turns & Actions
- Week 8: Win Conditions & Testing

Schedule & Milestones

Sprint 2

- Week 10: New Heroes
- Week 11: Team Selection
- Week 12: Abilities
- Week 13 Enemy Types & Encounters
- Week 14: Finishing Touches & Testing

Project Documentation

- Project Plan Presentation (PPP)
- GitHub Repository